

F.Y. BCA [Sem-1]

Subject : Internet & HTML

Subject code : CC-103

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Unit – 1

(Introduction to Internet)

Internet

What is internet ?

- The Internet is the global system of interconnected computer networks that uses the Internet protocol suite (TCP/IP) to communicate between networks and devices.
- It is a world wide interconnected system of thousands of computer networks, each network in turn links thousands of computers together.
- The Internet carries a vast range of information resources and services, such as the inter-linked hypertext documents and applications of the World Wide Web (WWW), electronic mail, and file sharing.
- It is a network of networks.
- It is a global network of billions of computers and other electronic devices.

- It is an ocean of information.
- With the Internet, it's possible to access almost any information, communicate with anyone else in the world, and do much more.
- Facilities you can access on the internet :
 - Email
 - Stay connected using social media
 - Shopping
 - Managing bank accounts
 - Pay your bills
 - Watch TV/OTT platforms and get entertained
 - Learn new skills

Evolution and history of internet

- The history of the Internet has its origin in the efforts to build and interconnect computer networks that arose from research and development in the United States and involved international collaboration, particularly with researchers in the United Kingdom and France.
- Internet began as ARPAnet (arround late 1960s).
- ARPA – Advanced Research Projects Agency
- Till late 1970s ARPAnet operated to provide connection facility to around a dozen of research sites.
- During next two decades, the network that evolved was primarily used by academic institutions, scientists and the government for research and communications.
- In 1992, when the U.S. Govt. Began pulling out of network management, and commercial entities offered internet access to the general public for the first time.
- This change marked the beginning of the internet's expansion.
- In india, internet was inaugurated on 15th Aug., 1995.

How does the internet works ?

- The Internet works through a **packet routing network** in accordance with the *Internet Protocol (IP)*, the *Transport Control Protocol (TCP)* and other **protocols**.
- A **protocol** is a set of rules specifying how computers should communicate with each other over a network.
- Data sent across the Internet is called a *message*. Before a *message* is sent, it is first split in many fragments called **packets**. These *packets* are sent independently of each other.
- **Packet routing network** : It is a network that routes *packets* from a source computer to a destination computer.

The Internet is made up of a massive network of specialized computers called *routers*. Each *router's* job is to know how to move *packets* along from their source to their destination.

A *packet* will have moved through multiple *routers* during its journey.

Internet addressing & DNS

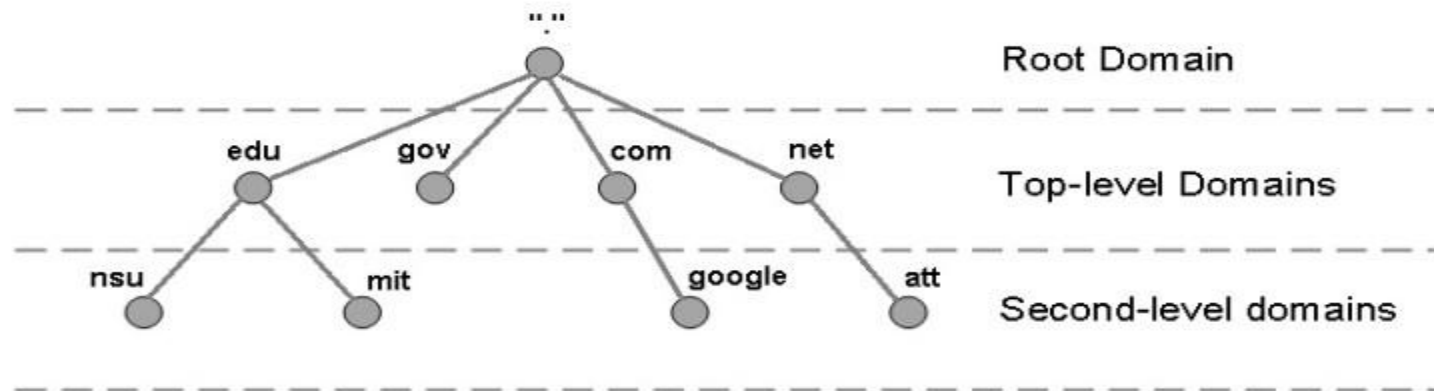
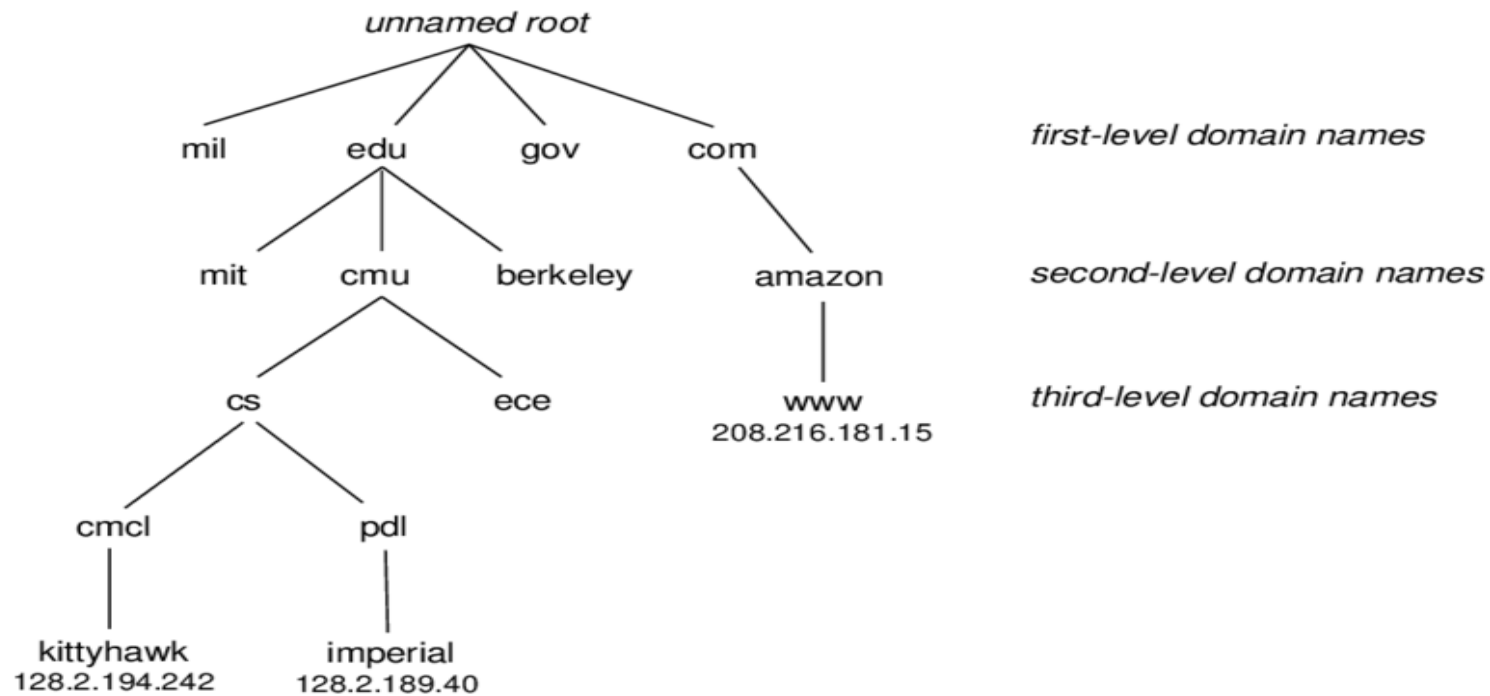
- You have to understand how the systems that are connected on the network are named and identified.
- With these names only you locate a computer and get connected to it.
- Every computer that is on the internet has its own unique address. This address is an electronic address.
- There are 2 kinds of addresses in the internet : (i) domain names (ii) IP addresses

Domain Name System and Servers

- On a network, each computer is identified by their IP address (eg : 192.168.10.234). Remembering the numbers is not easy but remembering names is much easier.
- So, a way was devised to associate an IP address with names that can be easily remembered. These names are called domain names.
- In simple words, domain name is your website's name.

- A domain name is the address where Internet users can access your website.
- A domain name is used for finding and identifying computers on the Internet.
- Domain name server is a distributed, scalable database of *IP addresses and their associated names*.
- To understand DNS and the way it is used, we need to understand the internet naming structure.

Eg: www.amazon.com



- Based on the top level identifications, there are basically 2 types of domain : (i) non-geographic domain (ii) geographic domain
- Non-geographic domains : com, edu, mil, gov, org, net
- Geographic domains : in, uk, pk, au, ca, jp, us

IP addresses

- Each host computer on the internet has a unique number called IP address. IP address uniquely identifies the computer in network.
- IPV4 IP address is a 32-bit number that is normally expressed in dotted-decimal format, with four numbers separated by dots, such as 192.168.123.132
- These numbers can be in the range of 0 to 255.

Internet V/S Intranet

- We have gone through the concepts and introduction of internet but what is this **Intranet?**
- Intranet is a private corporate network own by any private organization.
- In intranet, each computer is connected to the other computer for internal communication.

Internet	Intranet
It is worldwide / public network.	It is local / private network.
It has unlimited traffic .	It has limited traffic because of it's limited users.
It is not owned by any one entity.	It is owned by a specific organization.
It is managed by ISP (Internet Service Provider)	It is managed by network administrator who is working in an organization.
There are billions of servers operated by internet.	Only organization's server operated by intranet.
Internet has various LANs, MANs & WANs connected.	Intranet work on either LAN, MAN or WAN.

Switching

- A network is a set of connected devices.
- Whenever we have multiple devices, we have the problem of how to connect them to make one to one communication possible.
- One of the solution is **switching**.
- A switched network consist of a series of interlinked nodes, called switches.
- Switches are devices capable of creating temporary connections between two or more devices linked to the switch.

- 3 types of switching

1. Circuit switching

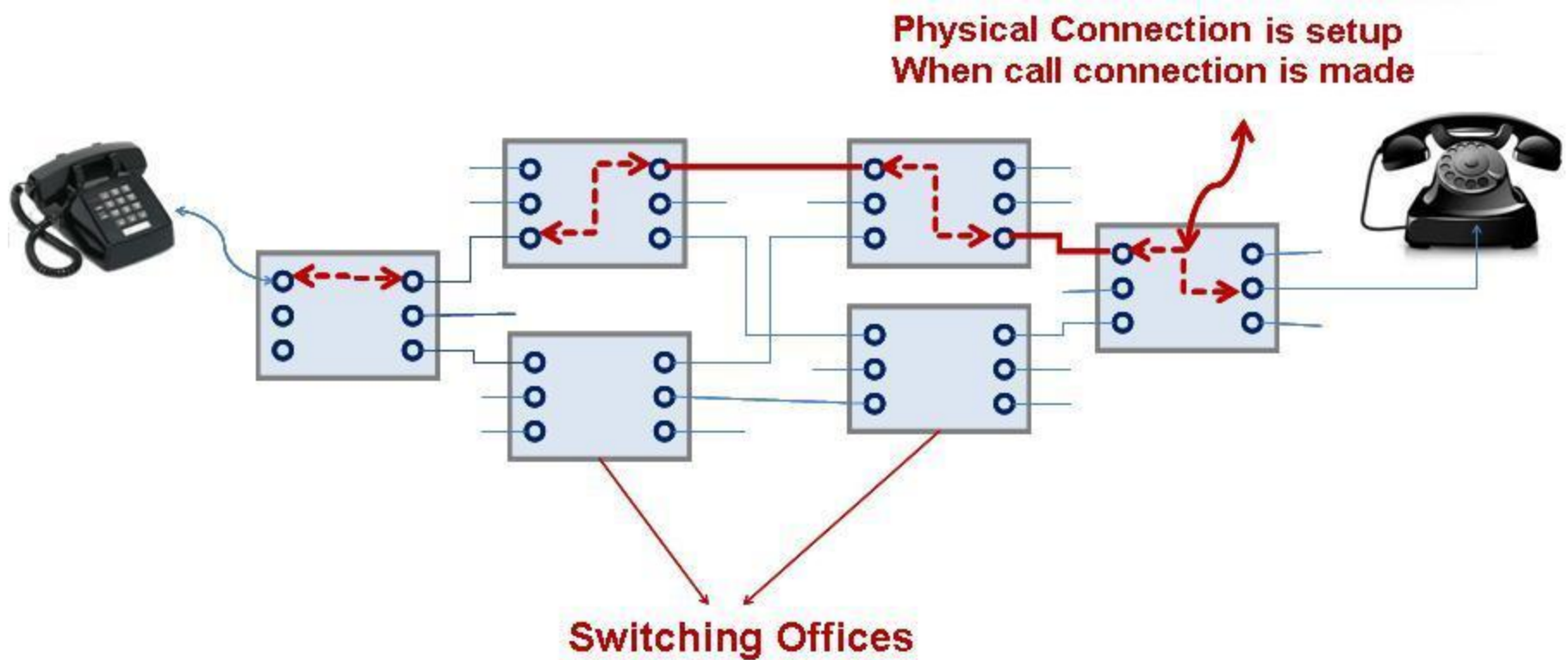
2. Packet switching

3. Message switching

1. Circuit switching

- It is the transmission technology that has been used since the first communication networks in the 19th century.
- Here, a sender must first establish connection to a sender before any communication is possible.

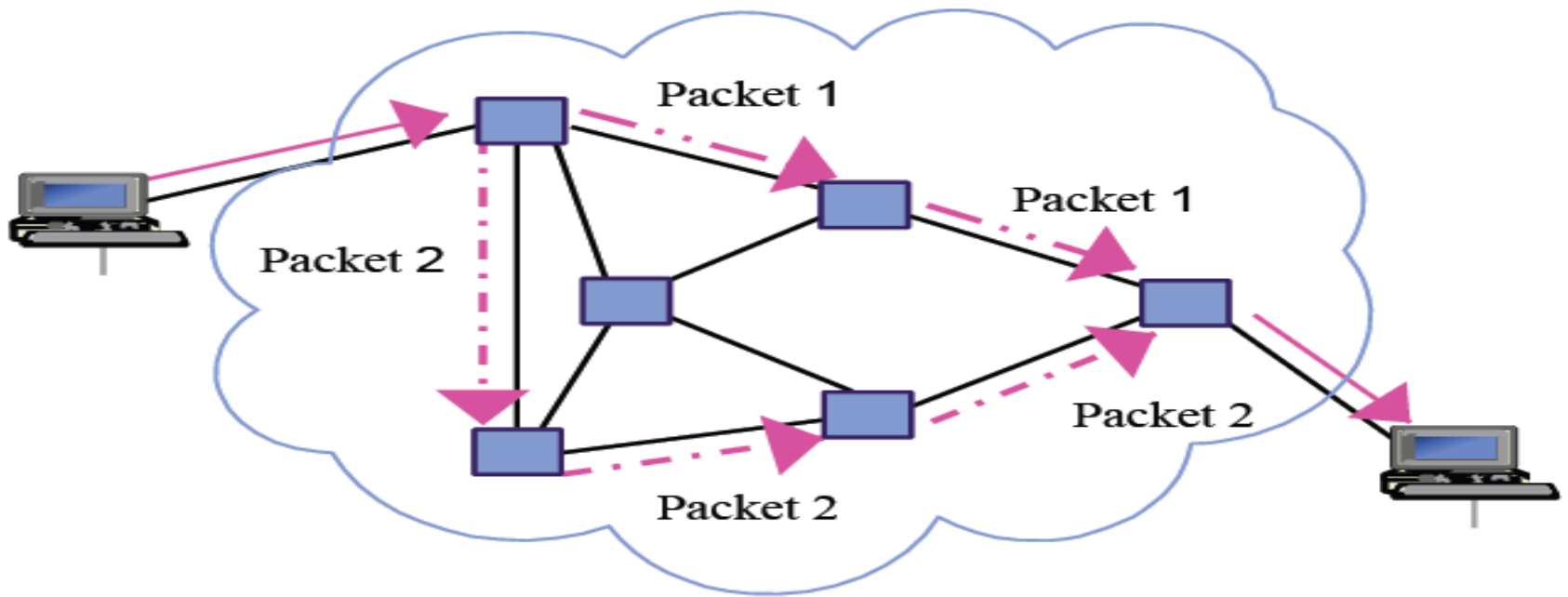
- The set of resources allocated for a connection is called a circuit.
- A path is a sequence of links allocated between nodes called switches.
- The path taken by data between its source & destination is determined by the circuit on which it is flowing, and does not change during the lifetime of connection.
- The circuit is terminated when the connection is closed.



2. Packet switching

- Packet switching is a method of grouping data that is transmitted over a digital network into suitably-sized blocks, called ***packets***.

- Packet switching is the primary basis for data communications in computer networks worldwide.
- The network over which packets are transmitted is a shared network which routes each packet independently.
- When a computer attempts to send a file to another computer, the file is broken into packets so that it can be sent across the network in the most efficient way. These packets are then routed by network devices to the destination.



3. Message switching

- There is no direct connection between the source and destination nodes.
- Here, intermediary nodes have the responsibility to convey the received message from one node to another.

- Here, messages are routed in their entirety, one hop at a time.
- Each message is treated as a separate entity.
- Each message contains addressing information, and at each switch this information is read and the transfer path to the next switch is decided.
- Each message is stored (on hard drive) before being transmitted to the next switch.
- Because of this it is also known as a 'store-and-forward' network.
- Email is a common application for message switching.

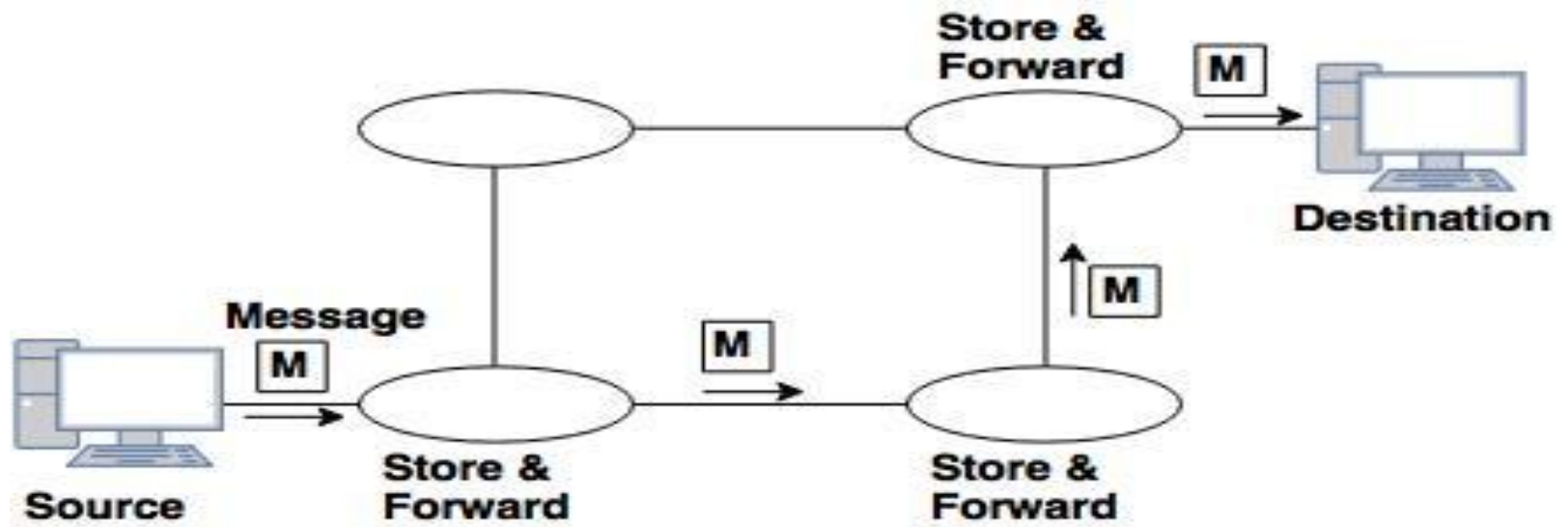


Fig: Message Switching

Types of connections

- Following are the different types of Internet connections :

1. Dial-Up connection
2. ISDN
3. ADSL
4. Leased line connections
5. Satellite connections

1. Dial up connections

- It is one type of connection available from ISPs.

- It is the slowest & cheapest connection.
- It allows us to connect to the internet via a local server using standard 56k modem.
- It establishes a connection to an Internet service provider (ISP) by dialing a telephone number on a conventional telephone line.
- Because telephone access is widely available, dial-up is often the only choice available for rural or remote areas, where broadband installations are not feasible.

- Broadband Internet access via cable, digital subscriber line, satellite has replaced dial-up access in many parts of the world.
- Pros
 - Economical, cost is very low
 - Hardware cost is nominal
 - Widely available
- Cons
 - Very slow
 - Same phone line can't be used for internet connection & for making a call
 - Increased Performance overhead

2. ISDN

- Integrated Services Digital Network
- It is an international communication standard for sending audio, video, data over a digital telephone lines OR normal phone wires.
- key feature : It integrates voice and data on the same lines.
- Two types of ISDN
 1. Basic rate interface (BRI)
 2. Primary rate interface (PRI)

Advantages :

- Digital service with less errors
- Direct fast connection with no dialing
- Higher bandwidth
- Streaming video OR audio without disruption.
- Supports multiple users
- Excellent quality

Disadvantages :

- More expensive than a standard telephone.
- Not easy to setup. Engineer has to install it.

ISDN services :

- Call forwarding
- Call waiting
- Three party conference
- Malicious call identification

3. ADSL connection

- Asymmetric digital subscriber line.
- It's a part of technologies based on the DSL concept.
- Allows faster data transmission.

- Supports 1.5 to 9 MBPS (Down-streaming) and 16 to 640 KBPS (Up-streaming).
- It requires special ADSL modem.

Advantages :

- Provides uninterrupted connections.
- Provides voice and data transmission simultaneously over same phone line.
- Faster speed, cheaper cost.

Disadvantages :

- Not available everywhere easily
- Special equipments have to be installed
- Network congestion and line noise are possible.

4. Leased-Line connections

- A **leased line** is a private bidirectional or symmetric telecommunication line between two locations.
- It's a permanent telephone connection between two points.
- It is typically used by business to connect geographically distant offices.
- Unlike dial-up connections, a leased line is always active.
- Leased lines can be used for telephone, Internet, or other data communication services.

- In leased line we don't need modem or ISDN terminal adapter to connect two systems.
- We have to pay fixed monthly rate as a Fee.
- It is also known as an Ethernet leased line, dedicated line, data circuit or private line.

Advantages :

- Provide secure and private dedicated connection.
- It can be laid for up to distance 35 km.
- Used for people requiring extra high bandwidth.
- Reliable and dependable.
- Provides high speed connection.

Disadvantages :

- Very expensive to install.
- Not suitable for single or home workers.

5. Satellite connections

- It is an arrangement where the outgoing and incoming data are sent from, and arrive at a computer through satellite.
- Internet access is provided through communications satellites.

- It is popular where services such as cable, DSL, ISDN etc are not available or expensive.
- It supports all type of traffic including VPN, VOIP and video conferencing.

Advantages :

- Economical for wide area connectivity.
- Cost of maintenance is lower than any other wired technology.
- Provides the facility to design a network without physical switches.

Disadvantages :

- Transmission problem can be there in foggy medium.
- Finite number of satellites can be placed in an orbit.
- Security problems are very high.
- Signals can easily be trapped.

Internet Service Provider (ISP)

- An **Internet service provider (ISP)** is an organization that provides services for accessing the Internet.
- Internet service providers may be organized in various forms, such as commercial, community-owned, non-profit, or otherwise privately owned.
- You have to subscribe to a provider for your internet connection.
- You use computer and modem to access the provider's system.

- We know many ISP organizations like, BSNL, Airtel, Reliance, etc.
- ISP provides wired as well as wireless connections.
- ISP also provides
 - Roaming facility (relevant for those who travel a lot)
 - Multiple login facility (more than one user can access internet connection simultaneously)

Computer Networks

- A computer network is a group of interconnected computers.

Objectives :

- To share resources such as database, software, etc.
- Distribution of processing & load sharing.
- Provides cheaper expandability.
- Provide a communication medium so that people sitting at geographically distant locations can exchange information.

❖ Uses of computer networks :

- Resources sharing
 - All the equipments and data are easily available to anyone at anytime on the network without the concern of location.
- Scalability
 - Performance is increased by adding more processors.
- High reliability
 - Several computers share various resources. In case of one resource failure, the other could be used .

- Saving money
 - Network consists of many small powerful computers with data kept on one or more shared file server machine which offers better price/performance ratio than mainframe computers.
 - This is achieved by client-server model.
- Powerful communication medium
- Access to remote location
- Person to person communication
- Interactive environment

- Interoperability
 - It is the ability of system to work with other systems without special efforts.
 - It is the ability of software and hardware on different machines from different vendors to share data.

❖ Network devices

- Network cable (twisted pair cable, coaxial cable or fiber-optic cable)
- Hub
 - cable interface units which sends and receives signals on the network cable.
- NIC
 - Network Interface Card is inserted into expansion slot inside the computer for attaching a cable to it.
 - Used to send and receive messages to and from the LAN.

- Router
 - Used to connect two or more similar type of LANs.
 - It acts like a bridge but for very large networks.
 - Also acts like a traffic controller.
- Bridge
 - It is a network device that connects two networks.
 - It links two networks.
- Gateway
 - Used to connect two dissimilar LANs.
 - It translates the LANs, different data formats.
 - Connects multiple networks & routes packets among them.

- Modems
 - Computers are **digital** machines. Any information that is sent to computers must be expressed in digits/numbers. They understand only binary numbers.
 - Telephone wires carry **analog** signals (they are electrical representation of sound waves).
 - So, Linking 2 computers with telephone system requires that both computers be equipped with **modem**.
 - Modem provides necessary translations between digital & analog worlds.
 - Modem = Modulator + Demodulator
 - Modulation (digital to analog translation)
 - Demodulation (analog to digital translation)

- 2 types of modems are there : internal & external
 - Internal : designed to fit within the computer's expansion slot
 - External : has its own case and power supply
- Modems can be classified on the basis of transmission speed
 - Low speed modems (up to 600 bps)
 - Medium speed modems (from 1200 to 1300 bps)
 - High speed modems (4800 bps to 19.2 kbps)

- Data multiplexers
 - Modems are intermediary devices used when the distance is large.
 - Another intermediary device is data multiplexer which allows sharing of transmission media.
 - Multiplexing is a technique for sharing a common channel between several users.
 - It reduces the cost of transmission media and modems.
 - Different multiplexing techniques are - space division multiplexing (SDM), frequency division multiplexing (FDM), time division multiplexing (TDM), etc.

❖ Network types

- 3 types of networks are there : (1) LAN (2) MAN (3) WAN
- **LAN (Local Area Network)**
 - LAN is a group of computers and associated devices that share a common communications lines.
 - It's a privately owned network
 - Covers small physical area like, office, single building or a campus up to few KM
 - Most common topologies are : Bus, Ring & Star

- **MAN (Metropolitan Area Network)**
 - Geographical scope is wider than LAN
 - It can connect different LANs residing in different geographical areas
 - Covers an area of between 5 and 50 KM
 - Can cover entire city or town
 - It can connect large networks through fiber optic, coaxial cable or in a wireless manner.
 - MANs are made from switches or routers connected to one another with high speed links.

- **WAN (Wide Area Network)**
 - It spans a large geographical area.
 - It connects a country or continent.
 - Communication is done by telephone, satellite, dishes and radio waves.
 - It covers both national and international locations.
- **Wireless network**
 - Telecommunication network which is connected without the use of wires.
 - Uses electromagnetic waves (i.e. radio waves)
 - **Adv** : convenience, mobility, expandability, cost saving
 - **Dis-adv**: security, insufficient range in large structure, slow speed

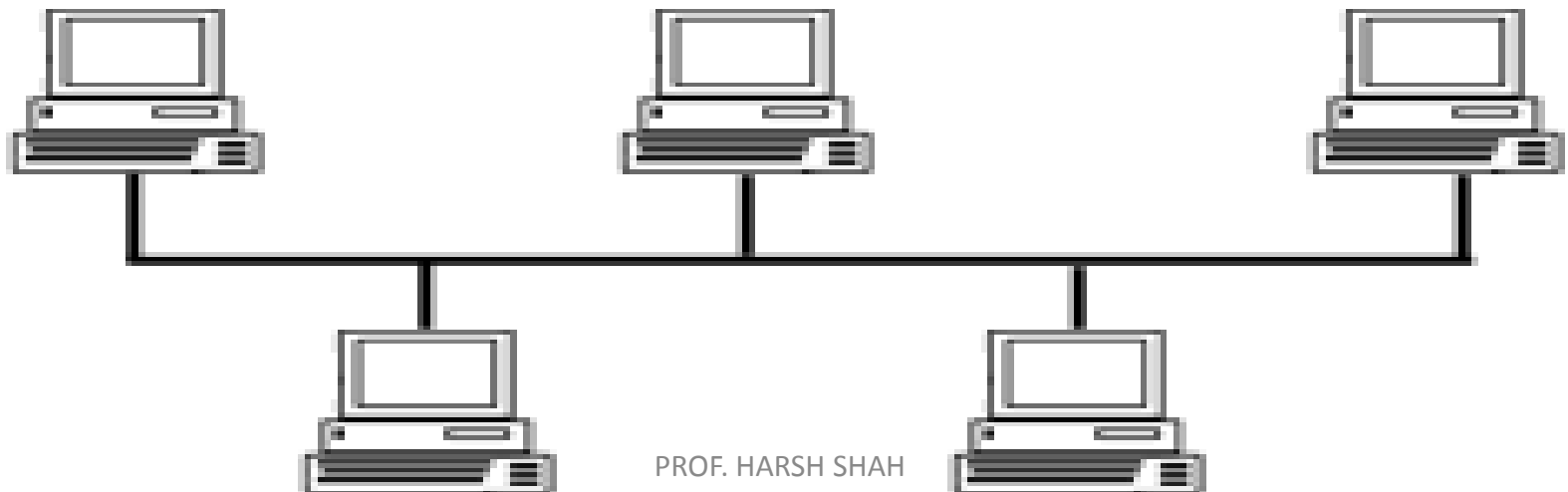
❖ Network Topologies

- It is a physical connection between each device in network.
- Topology shows relationship of devices in network.
- Following are the various topologies :
 - Bus topology
 - Star topology
 - Ring topology
 - Tree topology
 - Mesh topology

Bus topology

- Simplest topology
- Also called linear topology
- In this topology, all the nodes are connected to single cable (called bus – it is backbone of the network)

Bus Topology

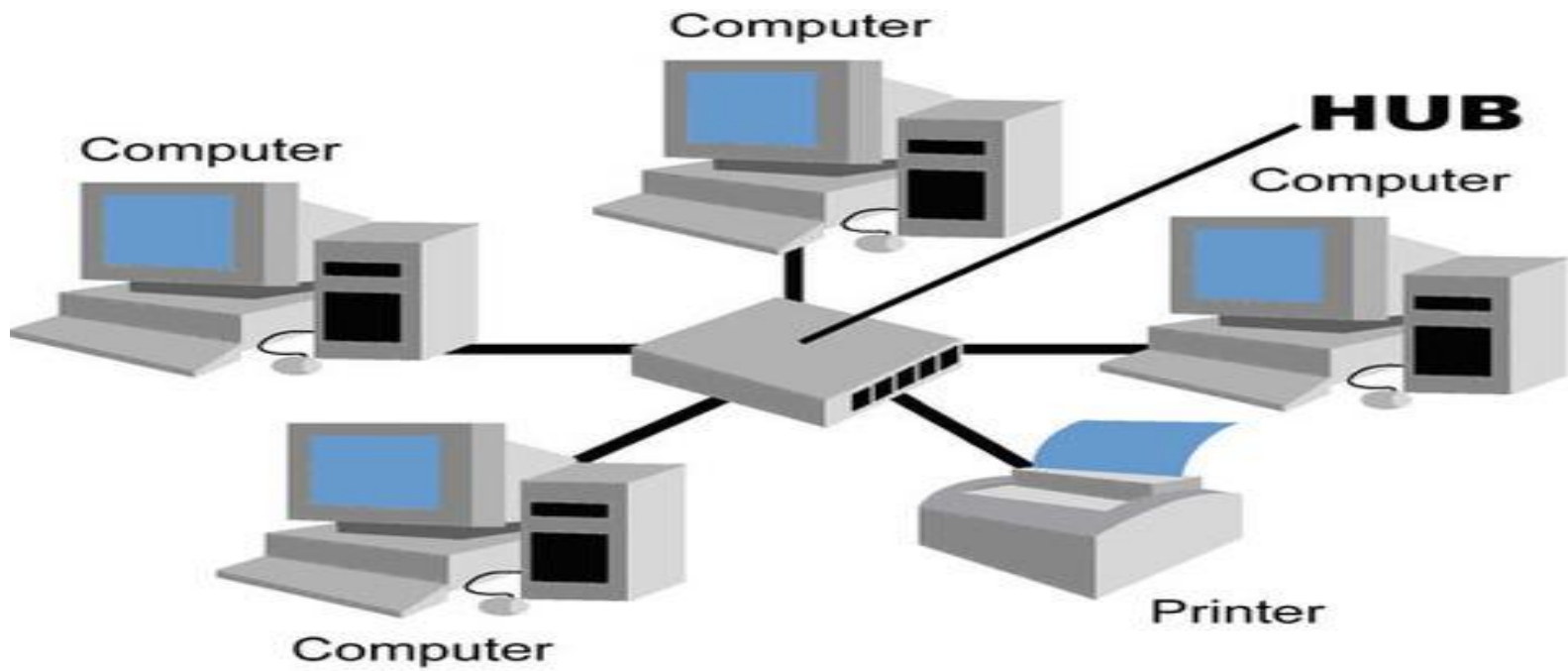


- Problem can be arise when two clients want to transmit at the same time on the same bus.
- For communication, a signal from the source node is broadcasted and it travels to all workstations, then the receiver (whose address is specified) accepts it.
- Advantages :
 - Easy and quick to implement as well as extend
 - Well suited for small or temporary networks not requiring high speed
 - Cost is very less as just a single cable is used
 - Cheaper than other topologies
 - Cable faults are easily identified
 - If one node (PC) fails, it does not affect the other computers

- Disadvantages :
 - Limited cable length and number of stations
 - Slower than other topologies
 - Requires more management
 - Problem in cable makes entire network down
 - Maintenance cost may be higher in long run
 - Heavy network traffic can slow down a bus
 - Low security as all the nodes receives the signal sent from source node.

Star topology

- All the components of network are connected to the central device called HUB.
- Each device has a dedicated link to connect to hub

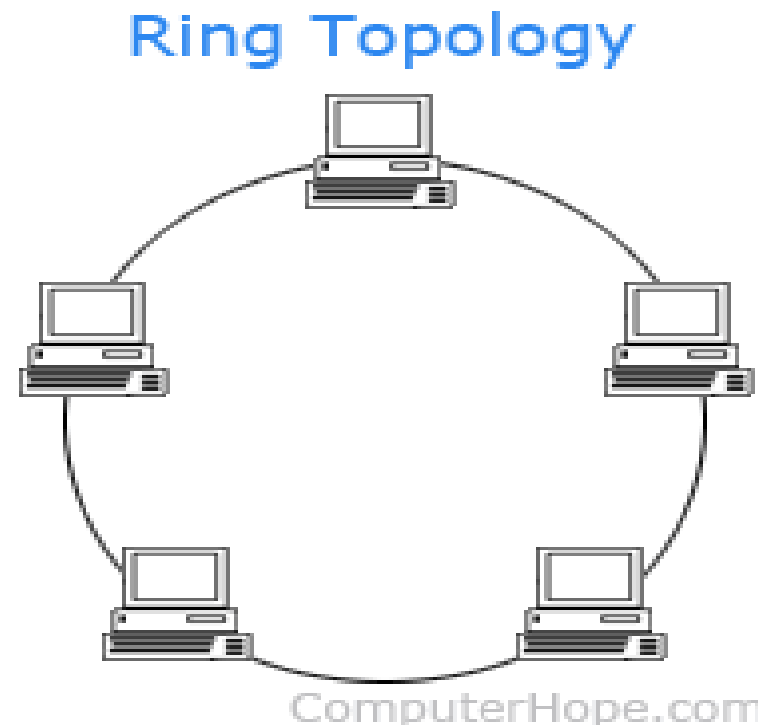


- Devices are not connected directly to each other.
- The data sharing from one computer to another is done only through Hub.
- Performance of the network is dependent on the capacity of central Hub.
- Requires more devices and cables to setup than a bus topology

- Advantages :
 - Better performance than bus topology
 - Isolation of devices
 - Centralized management (helps in monitoring the network)
 - Scalability is easy
 - Simplicity (easy to understand & establish)
 - Failure of one node does not affect the rest of the network
 - Easy to detect the failure and troubleshoot it
- Disadvantages :
 - High dependency on the hub
 - If the hub fails, entire network fails
 - Network size is limited by the number of connections that can be made to the hub
 - Performance depends on capacity of hub

Ring topology

- Here, each node connects to exactly two other node, forming a single continuous pathway for signals through each node – a ring.
- It's a kind of computer network configuration where each network computer / devices are connected to each other forming a large circle.
- Data travels from node to node
- It is hardly used now-a-days

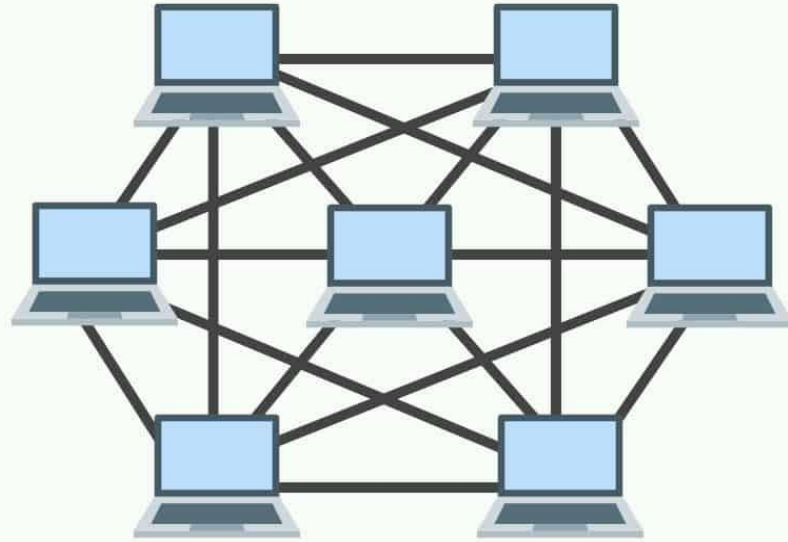


- Signals are passed along the ring in one direction from device to device until it reaches its destination
- Advantages :
 - Easy to install and reconfigure
 - Performance is better than a bus & star topology
 - Can create much larger network using token ring
 - Does not require network server to manage the connectivity between the computers
 - Easy to identify & isolate faults
 - Easy to add / remove new device in the network

- Disadvantages :
 - One malfunctioning workstation can create problems for the entire network
 - Bandwidth is shared on all links between devices
 - Moving, adding or changing the devices can disturb the network

Mesh topology

- Here, each of the computers / devices are interconnected with one another
- This topology is commonly used for wireless networks.
- A fully connected mesh network needs $\frac{n(n-1)}{2}$ physical channels to link 'n' devices.

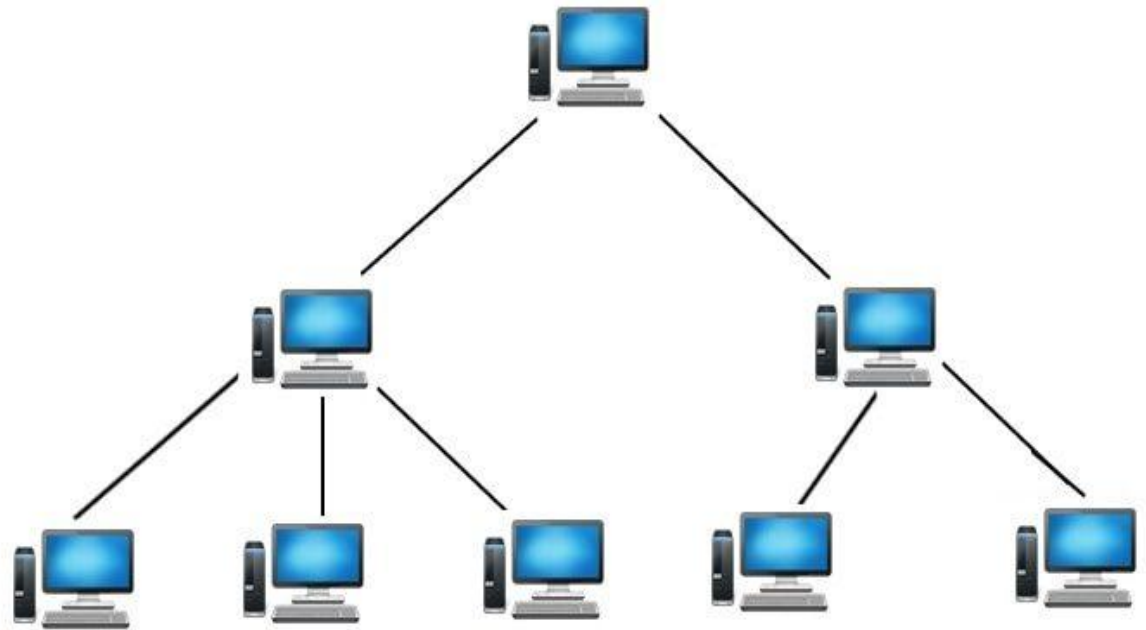


- Advantages :
 - Can handle high traffic easily
 - Network can be expanded without disruption to current users
 - Data transfer doesn't get affected as alternatives are always present in case of component failure
 - Identification and isolation of faults are easy

- Disadvantages :
 - Require more cables than the other topologies
 - Implementation is complicated
 - Expensive

Tree topology

- Also known as star bus topology
- It follows a hierarchical pattern
- It connects multiple star networks to other star networks
- Devices are connected to some of the secondary hubs, and secondary hubs are connected to the central hub



- Advantages :
 - It is supported by many network & hardware vendors
 - A point-to-point connection is possible with tree networks
 - One of the best topology for branched out networks
 - Easy to setup and extend the network
 - Easy to detect fault in the network

- Disadvantages:
 - If main central hub fails, the entire network fails
 - More expensive than other topologies
 - It's a big topology. So, it gets complicated after a certain point.

Hybrid topology

- It uses combination of two or more topologies
- It works only when two different topologies communicate together
- For Example : in a college, if different departments are having different topologies and if each department can communicate with each other, then this structure is called hybrid topology.

E-mail

- Electronic mail (e-mail) is a method of exchanging messages / mails between people using electronic devices.
- It is one of the basic and most popular internet service.
- When e-mail originated in 1970s, it was just possible to send textual message. But now we can attach documents, audio, video, photos, etc in the email.
- Sender can send an email to the receiver by specifying receiver's email Id.

E-mail system

- It is similar to our postal service.
- It works on internet so we need a web application to manage the mails. Eg : Gmail, yahoo mail, etc.
- You have to create a mail Id to access all the facilities given by these mail web applications.

Email service includes following **features & facilities** :

- Compose mail
- Send and receive mail
- Manage contacts in an address book
- Create a closed user groups - Search mails
- Attach documents - Save messages in a draft
- Send Carbon Copy (**CC**) and Blind Carbon Copy (**BCC**)

E-mail protocols

- Protocols are set of rules.
- E-mail Protocols are set of rules that help the client to properly transmit the information to or from the mail server.
- Following are different mail protocols.
 1. SMTP (Simple Mail Transfer Protocol)
 2. POP3 (Post Office Protocol 3)
 3. IMAP (Internet Message Access Protocol)
 4. MIME (Multipurpose Internet Mail Extensions)

1. SMTP

- SMTP is text based protocol, it only supports text.
- It is connection oriented, application level protocol.
- It handles exchange of messages between computer systems in the internet.
- It uses TCP (Transmission Control Protocol) which provides a reliable communication.
- Apart from transferring e-mail, SMTP also provides notification regarding incoming mail.
- It provides fast, efficient and reliable service.
- Drawback : it requires both nodes to be online for communication between them. That is why POP comes in.

2. POP3

- It is designed to allow single user hosts to read mail from a server.
- It is a mail collection and distribution system.
- It's a client server protocol in which email is received & held for you by your internet server.
- It allows creating a mailbox for each user who has a mail account on the server.
- Periodically, You have to check mailbox on the server and download the mails.
- Mail will be deleted from server as soon as you download it.
- 3 versions are there : POP, POP2, POP3 (latest version)

3. IMAP (Internet Message Access Protocol)

- Richer in functionality than POP.
- Allows hierarchical storage of mail.
- Allows client to access & manage mailboxes on the server as if they were local.
- Allows one user to access multiple mail servers and multiple users to share a single mailbox.
- IMAP is still used extensively, but is less important now that so much email is sent via web-based interfaces such as Gmail, Hotmail, Yahoo Mail, etc.

4. MIME (Multipurpose Internet Mail Extensions)

- It was designed as an extension to the SMTP allowing people to send binary data.
- Allows users to exchange different kinds of data files over the Internet such as images, audio, and video.
- Now MIME types are not just for use with e-mail; they have been adopted by Web servers as a way to tell Web browsers what type of material was being sent to them so that they can cope with that kind of messages correctly.

Email addresses

- An email address identifies an email box to which email messages are delivered.
- Example : harsh@gmail.com
- Here, the part before @ sign (harsh) is a user-id that you select and the part after @ sign (gmail.com) is the server that stores the email addresses.

Structure of email messages

- Email message has 3 parts
 1. Mail headers
 2. Message body
 3. signature

1. Mail headers

- It is a piece of information that tell you a number of things about a particular piece of email.
- Headers has specific name and a specific purpose.
- Different header type : from, date, to, cc, subject, reply-to, etc

from: **Aanal Desai** <aanaldesai@cpi.edu.in>
to: Harsh Shah <harshshah@cpi.edu.in>,
Director BCA <directorbca@cpi.edu.in>
date: Jul 15, 2019, 12:02 PM
subject: Lecture adjustment for 22nd July,2019
signed-by: cpi-edu-in.20150623.gappssmtp.com
security:  Standard encryption (TLS) [Learn more](#)
 : Important according to Google magic.

2. Message body

- This is the part of message where you type actual text.
- You can format it with colors, pictures and pre-define styles.

3. Signature

- It's a sequence of lines usually giving some information about the person who has sent the email. It's optional.
- Usually a signature has sender's full name, designation and some contact information.

FYBCA Improvement Test (Theory/Practical) March 2019 Schedule and Candidate List



Harsh Shah <harshshah@cpi.edu.in>

Sat, Mar 2, 4:26 PM



to Director, Rutvi, Aanal, Nirav, Parimal, Shivang, krutijani ▾

Respected All,

Kindly verify the attached schedule & list of candidates (FYBCA SEM-II) for Improvement/Retest. Please let me know if any query is there.



**Message
Body**

--
Harsh Shah

Asst. Professor

SCPICA (BCA Dept.), Gujarat University.

Signature

Email clients and servers

- Email clients : The software that you install on your PC to send and receive is called email client.

Windows has its own email client called – Microsoft Exchange. MS Office comes with Microsoft Outlook that can also send and receive mail.

- Email servers : Also known as MTA (mail transfer agent). It receives incoming email from users and forwards outgoing email for delivery.

A computer on which such applications are running is also called a mail server.

Mailing List

Emails lets you manage the mailing lists.

You can send mails to a large group.

Sending message to a mailing list is called **Posting**.

- Moderated and unmoderated mailing list
 - Moderated list : here, each message is first read by a moderator who decides if the message is appropriate for group or not, then appropriate message is sent in the group.
 - Unmoderated list : here, all messages are automatically sent to everyone in the group.

- Mailing list administration
 - Keeping track of the people on the list
 - People can subscribe/unsubscribe to the mailing list
 - Mailing lists are administered in 2 ways :
 - (i) Manually maintained list (managed by administrator who keeps track of every member)
 - (ii) Automated list (managed by a program that keep track of every member)
- Mailing list management programs
 - Manages the mailing lists
 - Popular programs : LISTSERV, ListProc

Email security

- It deals with 2 issues : safety of the information that you send **AND** reliability of the information that you receive
- Email security is required to filter the spam data and send them to spam list.
- It is also used to hide sender and receiver email addresses from the hackers OR sniffer programs.

World Wide Web (WWW)

- www is an internet browsing and information retrieval system that helps you to find and retrieve information, using links to other www pages.
- It provides a network of interactive documents and the software to access them.
- Basic elements of www : (explain)
 - Client / server system : web is based on client/server system. Client sends a request for some resource. Server accepts the requests and provides the requested resource in response.

- HTTP : Hyper Text Transfer Protocol is a standard protocol that allows web clients and servers to communicate. It carry a request from a browser to web server.
- URLs : used to locate internet resources (i.e. web pages)
- HTML (Hyper Text Markup Language) : it's a universal language of web. Web pages are developed in HTML.
- Multimedia : it includes graphics, sound, video and animation which is widely available over the web.

Search Engines

- A **web search engine** or **Internet search engine** is a software system that is designed to carry out **web search (Internet search)**.
- It is useful to search the World Wide Web in a systematic way for particular information.
- You enter list of keywords (query) to search something, and Search engine returns you list of web pages that contain those keywords/phrases.
- Search engines are databases that contain references to thousands of resources.

- Some of popular search engines are :
 - Google
 - Yahoo
 - AltaVista
 - Bing
 - Ask.com

Search criteria

- Basic search :
 - Basic search is done using keywords. A user types a word or words called “keyword” that he would like to search.

- Advanced search :
 - SE also provides an option of advanced search
- Case sensitivity :
 - It is the ability of SE to distinguish between upper case and lower case letters.
 - Most search engines are NOT case sensitive.
- Phrases :
 - Are group of words user enter to search.
 - Eg : “PubG new version”
- Required terms :
 - Specified with + symbol.
 - Eg : + powerbuilders

- Prohibited terms :
 - The terms you don't want to include in your search result.
- Wildcards :
 - *, ?, # are some of the wildcard characters
 - Eg : using **power*** as a search keyword will give you the results like powerful, powerhouse, powersoft, etc
- Boolean operators :
 - AND, OR, NOT are the Boolean characters we can use in a search phrase.
- Proximity searching
 - Proximity operators (eg. NEAR) help us to search for words within a certain distance of one another in database.

Web search agents

- Also called a **spider** - it is a program that visits Web sites and reads their pages and other information in order to create entries for a search engine index.
- The major search engines on the Web all have such a program, which is also known as a "crawler" or a "bot."
- Google's spider name is – Googlebot.

Unit – 3

(Working with Text, List, Tables and Frames)

Working with TEXT

Adding plain text to an HTML page

- We can add plain text in an HTML page between `<body>` and `</body>` tags.

- Eg :

```
<html>
```

```
  <head> </head>
```

```
  <body>
```

```
    Hello World
```

```
    This is HTML 5
```

```
  </body>
```

```
</html>
```

Adding text in new line

- We can use `<br>` tag for Adding text in new line in HTML document.
- Eg :

```
<html>
```

```
  <head> </head>
```

```
  <body>
```

```
    This is line-1 <br>
```

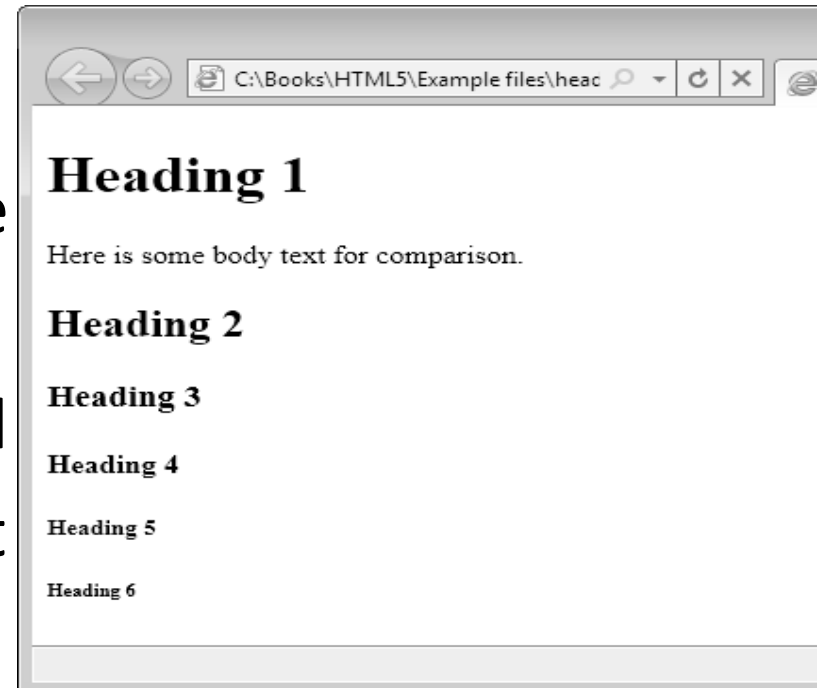
```
    This is line-2
```

```
  </body>
```

```
</html>
```

Creating headings on web page

- Headings in Web pages function the same way as they do in printed documents—they separate text into sections.
- The HTML standard defines six levels of headings, `<h1>` through `<h6>`, each one progressively smaller in font size.
- `<h1>` - biggest heading in size
- `<h2>` - smaller than `<h1>` in size
- `<h6>` - smallest in size
- Headings always looks bold and dark, and the default alignment is in left.



- Eg :

```
<html>
```

```
  <head> </head>
```

```
  <body>
```

```
    <h1> Heading 1 </h1>  <br>
```

```
    <h2> Heading 1 </h2>  <br>
```

```
    <h3> Heading 1 </h3>  <br>
```

```
    <h4> Heading 1 </h4>  <br>
```

```
    <h5> Heading 1 </h5>  <br>
```

```
    <h6> Heading 1 </h6>  <br>
```

```
  </body>
```

```
</html>
```

Creating paragraph

- We can use `<p>` tag.
- It will automatically break line of your paragraph. It will ignore extra spaces given in paragraph text and display only single space in between the words.
- Align attribute can be used to adjust the alignment (left, right, center, justify) of paragraph.

- Eg :

```
<body>
```

```
  <p> This is a paragraph.
```

```
    It contains multiple lines and    spaces.
```

```
    Write it as it is and check the output.  </p>
```

```
</body>
```

Creating a horizontal ruler

- `<hr>` tag is used.
- It draws a horizontal line in webpage.
- It is a void element. So, it must have start element `<hr>` but not necessary to write end tag `</hr>`.
- Eg :

```
<body>
```

```
  Hello <br>
```

```
  <hr>
```

```
  How are you ?
```

```
</body>
```

- Eg with attributes :

```
<hr color="red" align="center" width="30px" size="50px">
```


Creating subscript and superscript

- Superscript formatting makes text smaller and raises it off the *baseline*. `<sup>` tag is used. Eg : 1st, 2nd, etc
- Subscript makes text smaller and lowers it below the baseline. `<sub>` tag is used. Eg : H₂SO₄
- Eg :

`<body>`

1`<sup>`st`</sup>`, 2`<sup>`nd`</sup>`

`<br>`

H`<sub>`2`</sub>`SO`<sub>`4`</sub>`

`</body>`

Aligning the text

- `align` attribute is used.
- Attributes are used inside the html tags.
- Attribute value can be – left, right, center, justify
 - Left : align text in left side
 - Right : align text in right side
 - Center : align text in center
 - Justify : justify the text by setting indents from first line
- Eg :
`<p align="center"> text </p>`
`<h2 align="right"> text </h2>`
- `<center> </center>` tag can also be used to align the text in center.

Formatting the text

- Bold : `<b>` tag is used to make the text bold (dark).
Eg : `<b> This is bold text </b>`
- Italic : `<i>` tag is used to mark the text in italic style.
Eg : `<i> This is italic text </i>`
- Underline : `<u>` tag is used to mark the text in underlined style.
Eg : `<u> This is underlined text </u>`

Grouping the text

- By grouping, we can also use more than one tag inside single tag / single section.
- It is used to give common style and formatting to all the elements.

- `<div>` and `<span>` tags are used.
- These tags create a section / division in which we can put the content.
- `<span>` tag can also be used to apply inline style.
- Eg :
- `<div style="background-color:grey;">`
`<h3> This heading is inside the DIV tag </h3>`
`<p> This is a paragraph inside the DIV tag </p>`
`</div>`
- `<p> The span tag
 and the div tag
 are used like this.</p>`

Indenting Quotations

- You can indent the text as quotations.
- `<blockquote>` tag is used for long and multiple quotations. You don't need to specify the margin, this tag will automatically indent the text.
- `<q>` tag can be used for shorter quotations. The text written inside this tag will be surrounded by quotation marks ("").
- Eg :

`<blockquote>` write multiple lines here `</blockquote>`

`<q>` look, text is surrounded by quotation marks `</q>`

Working with character entities (special characters)

- The special characters are referred as entities. We can't access special characters in HTML. To access the special characters, we have to use it's code.
- i.e. we have to write the code to use copyright, trademark, less than (<), greater than (>)

Symbol	Entity Name	Entity Number
& (ampersand)	&	&
< (less than)	<	<
> (greater than)	>	>
(nonbreaking space)	 	
¢ (cent)	¢	¢
£ (pound)	£	£
¥ (yen)	¥	¥
© (copyright)	©	©
® (registered trademark)	®	®
° (degree)	°	°
± (plus or minus)	±	±
† (dagger)	†	†
™ (trademark)	™	™

- Eg :

<p>I will display €</p>

<p>I will display &</p>

<p>I will display <</p>

Commenting the text

- Comment tags are used to insert comments (additional, extra information) in the HTML source code.
- Commented text will not appear on a web page, it will not affect the design or layout of the web page.
- Opening tag : <!--
- Closing tag : -->
- Eg : <!-- This is the comment -->

Working with Lists

- Sometimes in a webpage, you need to display items in the form of list.
- List makes your webpage more structured.
- List can also be nested within another list.
- 3 types of list are there :
 - Ordered list
 - Unordered list
 - Definition list

1. Ordered list :

- It is a numbered list used to display list of items that need to be placed in a specific order.
- Ordered list can be displayed with numbers, alphabets and roman numbers.
- `<ol>` and `</ol>` tags are used to specify the list.
- `<li>` and `</li>` tags are used inside `<ol>` tag to specify the list items.
- We can use TYPE attribute to specify the type of number / alphabet displayed before your list.
(type="i" OR type="A" OR type="a" OR type="1")

Eg : `<ol>`

`<li> One </li>`

`<li> Two </li>`

`<li> Three </li>`

`</ol>`

`<ol type="a">`

`<li> Apple </li>`

`<li> Banana </li>`

`<li> Cherry </li>`

`</ol>`

2. Unordered list

- `<ul>` and `</ul>` tags are used.
- It displays the items with a bullet sign. We can select different types of bullets using the TYPE attribute.
- You can use this tag when you want to set an item list in any order & items need no numbering.
- We can use the TYPE attribute to specify the type of bullet displayed before your list. There are 3 types of bullets – disc, square & circle.

Eg : `<ul>`

`<li> Google </li>`

`<li> FB </li>`

`<li> Insta </li>`

`</ul>`

`<ul type="square">`

`<li> Amar </li>`

`<li> Akbar </li>`

`<li> Anthony </li>`

`</ul>`

3. Definition list :

- It is a list of terms and their definitions.
- It displays elements in definition form like in dictionary.
- Web browser indents each definition on the next line of the defined term.
- `<dl>` - used to create definition list
- `<dt>` - used to describe a term in a list
- `<dd>` - used to describe definition in a list

Eg : `<dl>`

`<dt> Coffee </dt>`

`<dd> It is a Black hot drink </dd>`

`<dt> Milk </dt>`

`<dd> It is a White cold drink </dd>`

`</dl>`

Output :

Coffee

Black hot drink

Milk

White cold drink

Nested Lists :

- Nested list is a list within the list.
- An individual list item can contain another entire list, called a nested list.
- Using this concept, you create one list inside another list.
- It is usually used to display sub items / sub sections.
- You can use both the ordered & unordered list as a nested list.
- Nested lists are quite useful, and often form the basis for navigation menus, as they are a good way to define the hierarchical structure of the web site.

Eg :

```
<ol>
```

```
<li>Chapter One
```

```
<ol type="a">
```

```
<li>Section One</li>
```

```
<li>Section Two </li>
```

```
<li>Section Three </li>
```

```
</ol>
```

```
</li>
```

```
<li>Chapter Two</li>
```

```
<li>Chapter Three </li>
```

```
</ol>
```

Output :

1. Chapter One

a. Section One

b. Section Two

c. Section Three

2. Chapter Two

3. Chapter Three

Working with Frames

- HTML frames are used to divide your browser window into multiple sections where each section can load a separate HTML document.
- Using frames, you can view data of all these documents simultaneously in a single web page.
- A collection of frames in the browser window is known as a frameset.
- The window is divided into frames in a similar way the tables are organized - into rows and columns.

Creating Frames

- Earlier we were using `<frameset>` & `<frame>` tags in HTML4. But in HTML5, these tags are replaced with the new tag - `<iframe>`.
- `<iframe>` tag is used to specify an inline frame.
- It is used to embed another html document within main HTML document.
- Using this tag will define a rectangular part in web page with border and scrollbar to define another document in main document.
- Attributes : src, name, height, width, sandbox

- src : set path of another html doc
- name : it gives name to iframe
- width & height : specifies frame's width & height
- sandbox : used to provide restriction to the content such as disabling scripts or plug-in given in the internal frame
- sandbox attribute takes the following **values** :
 - allow-same-origin : disables the plug-in that are generated in the frame content.
 - allow-forms : disables the forms given in the internal frame.
 - allow-scripts : refers to the values of internal element. This value disables the script to be run.

Eg :

```
<body>
```

```
  <iframe src="http://www.google.com"> </iframe>
```

```
  <iframe src="page2.html" width="150" height="150">  
  </iframe>
```

```
</body>
```

Applying hyperlink targets to a frame

- Hyperlink is used to create links between web pages.
- We have to use “href” attribute inside `<a>` tag.
When user clicks a link, the referenced page opens in another frame (on same page) OR in another document.
- “target” attribute is used to specify the target location of the web page.

Using named frame as hyperlink targets

- Named frames are frames for which you assign a **name**. You have to just pass the name of frame to the “target” attribute. When you click the hyperlink, the referenced page opens in the **target** frame.

Eg :

Create 1st page : frame.html

```
<iframe src="mainpage.html"> </iframe>
```

```
<iframe src="defaultpage.html" name="display">
```

```
</iframe>
```

Create 2nd page : mainpage.html

```
<body>
```

```
    <a href="page1.html" target="display"> page1 </a>
```

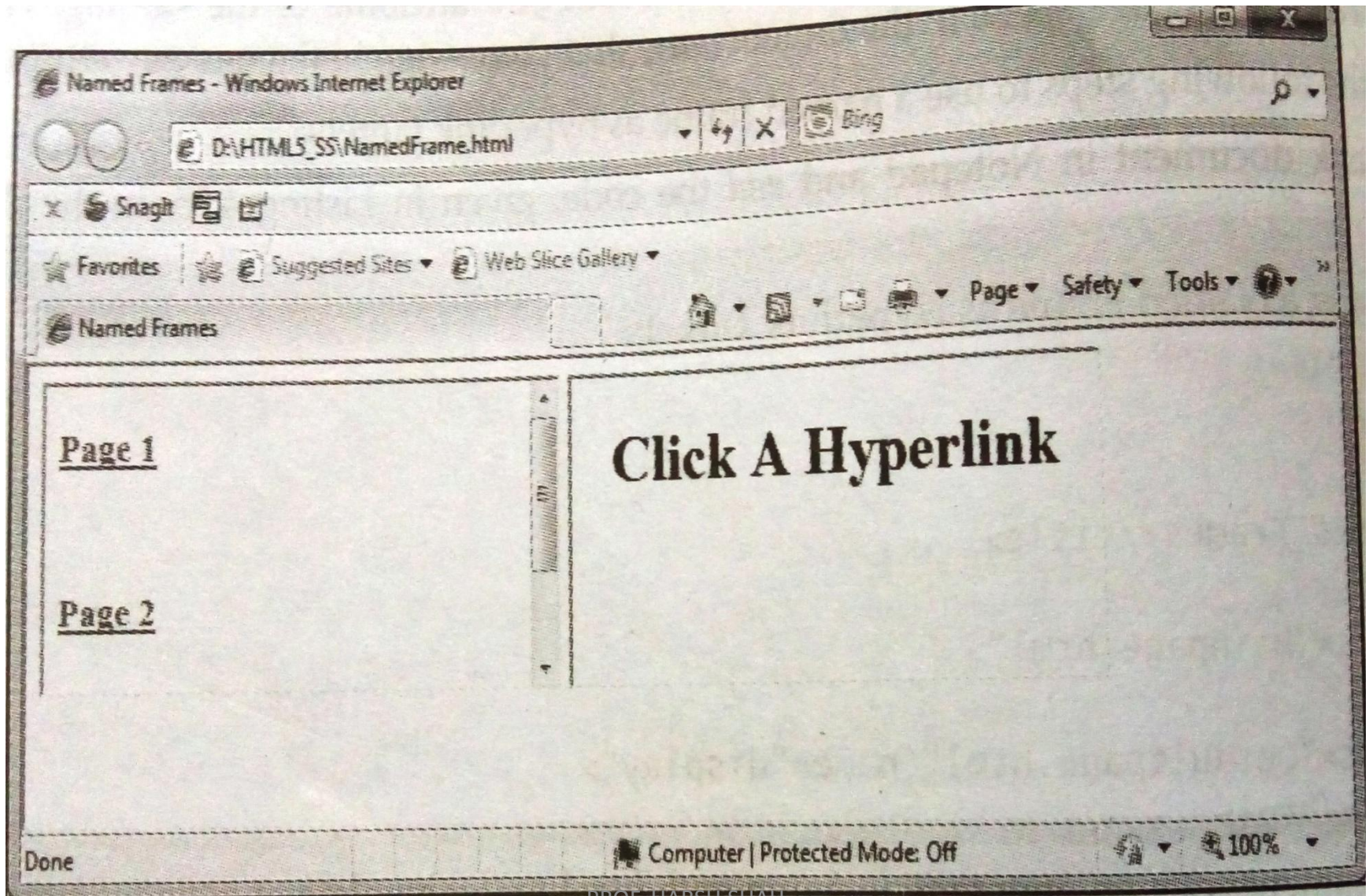
```
    <a href="page2.html" target="display"> page2 </a>
```

```
</body>
```

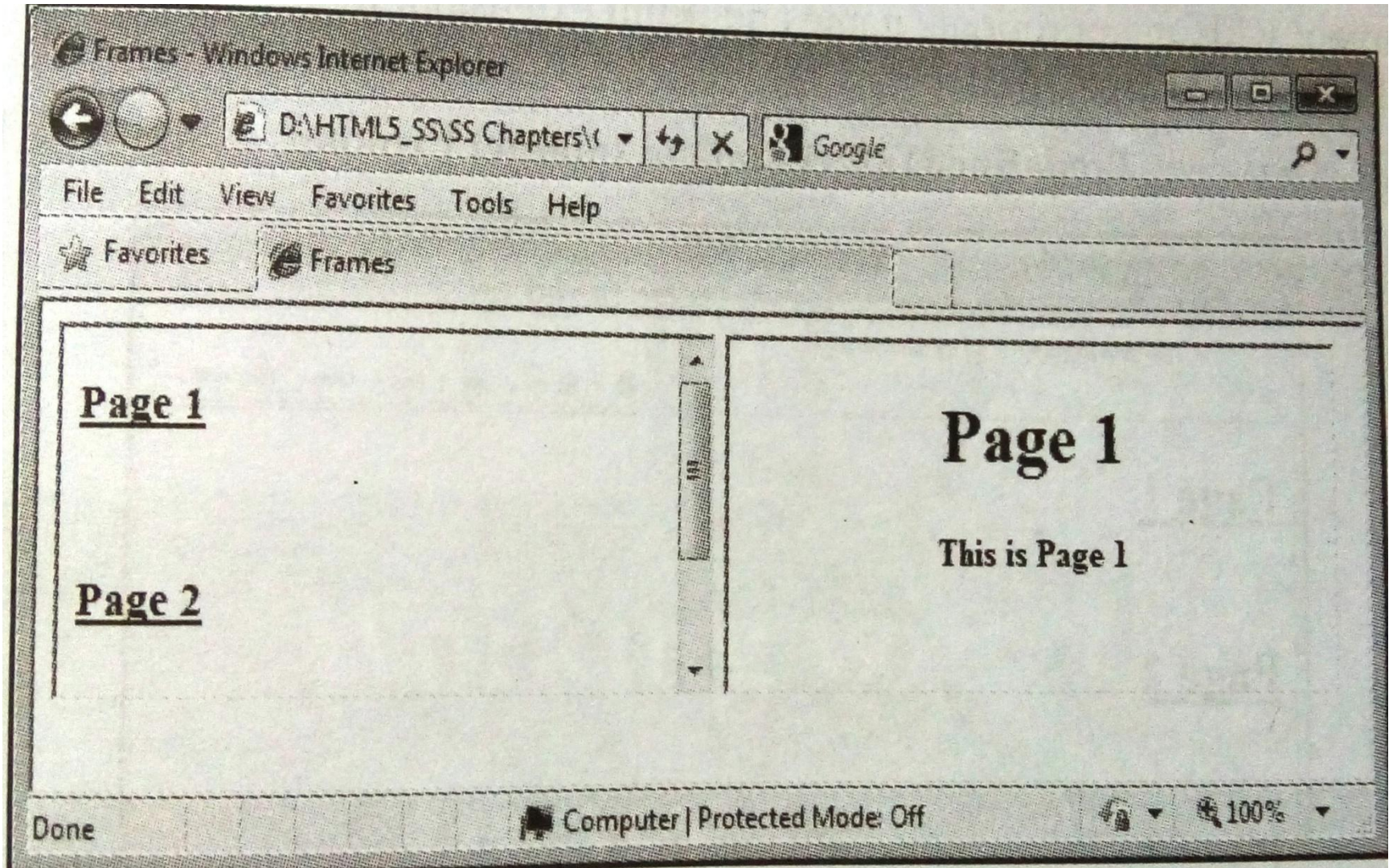
Create 3rd page : defaultpage.html – write anything (welcome message) in it.

Create 4th & 5th page : page1.html & page2.html

frame.html



Mainpage.html



Using a predefined target name as hyperlink targets

- Another option is to use predefined target names instead of specifying name of the frame to the “target” attribute.
- Predefined target names take one of the 4 values :
_blank, _top, _self, _parent.
- If you use _blank, the webpage opens in a new blank window.

Eg :

```
<body>
```

```
  <a href="page1.html" target="_blank"> page 1 </a>
```

```
  <a href="page2.html" target="_blank"> page 1 </a>
```

```
</body>
```


Working with tables